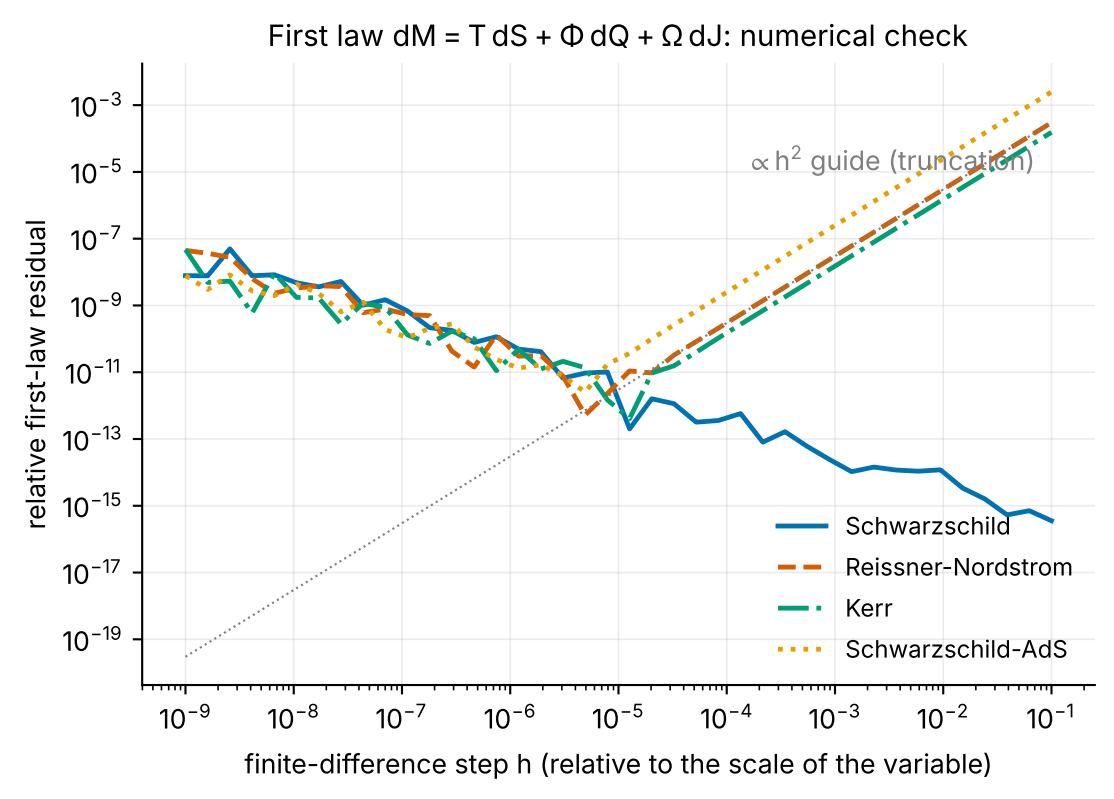
First law dM = T dS + Φ dQ + Ω dJ: numerical check
relative first-law residual
finite-difference step h (relative to the scale of the variable)
∝ h² guide (truncation)
Schwarzschild
Reissner-Nordstrom
Kerr
Schwarzschild-AdS
10⁻³
10⁻⁵
10⁻⁷
10⁻⁹
10⁻¹¹
10⁻¹³
10⁻¹⁵
10⁻¹⁷
10⁻¹⁹
10⁻⁹
10⁻⁸
10⁻⁷
10⁻⁶
10⁻⁵
10⁻⁴
10⁻³
10⁻²
10⁻¹